

Guardians of the Forest

Official Game Document

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Elevator Pitch

Guardians of the Forest is a 2D fantasy puzzle-adventure game inspired by the core mechanics of the popular *Angry Birds* franchise, where players fire magical projectiles from one of three unique wizards to destroy obstacles and defeat enemies, becoming a true Guardian of the Forest.

Table of Contents

Elevator Pitch	1
Table of Contents	2
Game Concept	3
Game Summary	3
Genre & Target Audience	3
Target Platform	3
Unique Selling Points	3
Story & Setting	4
Premise	4
World & Environment	4
Characters	4
Narrative Flow	7
Gameplay Overview	8
Core Mechanics	8
Objectives	8
Rules & Systems	8
Progression	9
Level Design	10
Level Layout	10
Challenges & Puzzles	10
Rewards & Progression	10
Art & Visuals	11
Art Direction	11
Characters & Environment	11
HUD & UI Elements	11
Animations	15
Audio	17
Music	17
Sound Effects	17
Controls & User Interface	18
Controls	18
UI Flow	18
Technical Specifications	19
Engine & Tools	19
Platform Requirements	19
Performance Targets	19
Development & Testing Hardware	20
Primary Development Machine	20
Quality Assurance Testing	20
Postmortem	21
Future Plans	23
References & Inspirations	24

Game Concept

Game Summary

Guardians of the Forest is a fantasy-themed, puzzle-adventure game where players select one of three wizards – each with their own unique projectile selection – to battle through eight enemy-filled levels to become a true Guardian of the Forest. The game adapts the well-known “pull-and-launch” gameplay of the popular *Angry Birds* franchise, but infuses it with magical spells, mystic environments, and a family-friendly setting.

The game was developed as my first full solo project with all programming, game design, audio editing, asset acquisition, and testing completed independently. The primary goal of the project was to gain hands-on experience with the full game development cycle.

Genre & Target Audience

Guardians of the Forest is a 2D single-player, puzzle-adventure game aimed at a wide and casual audience. Players who enjoy physics-based puzzles experiences, magical fantasy themes, or any title from the *Angry Birds* franchise will be sure to enjoy what's in store.

Recommended age: **PEGI 3** – Mild violence.

Target Platform

The game targets Windows PC exclusively. Limited experience with mobile and console development defined a scope centred on the Unity PC build pipeline.

Unique Selling Points

- **Unique Take on a Classic Formula:** Mixes the classic *Angry Birds* formula with a magical, fantasy twist.
- **Interesting Replayability:** With three unique wizards who each own two unique projectiles; there is plenty of replayability within the game.
- **Unlimited Target Audience:** The game is suitable for almost all ages with family-friendly visuals, gameplay, and themes.
- **Short But Sweet:** The small nature of the game ensures play sessions are satisfyingly short but sweet for players.

Story & Setting

Premise

The mystical forest of the Guardians has been seized by evil skeletons, goblins, and golems. As one of the three Guardian Wizards – each with their own unique magical projectiles – you must push through enemy-occupied forest regions and reclaim the land, becoming a true Guardian of the Forest in the process.

World & Environment

The game takes place across various mystical forest environments, each themed to match the encountered enemies. The tone is light, colourful, and accessible, blending whimsical fantasy with mild elements of danger.



Figure 1: The world environment of Guardians of the Forest.

Characters

Guardian Wizards (Protagonists)

Elemental Wizard

Bio: Focuses on bending Earthen elements to their will to destroy their enemies.

Projectiles: **Fireball**, **Electricball**

Playstyle: High-impact, destructive spell casting with visually explosive power.



Figure 2: The Elemental Wizard.

Healer Wizard

Bio: Focuses on their medicinal skills to defeat enemies standing in their way.

Projectiles: **Bloodball, Poisonball**

Playstyle: Pre-planned, delayed interactional spell casting with devastating influence.

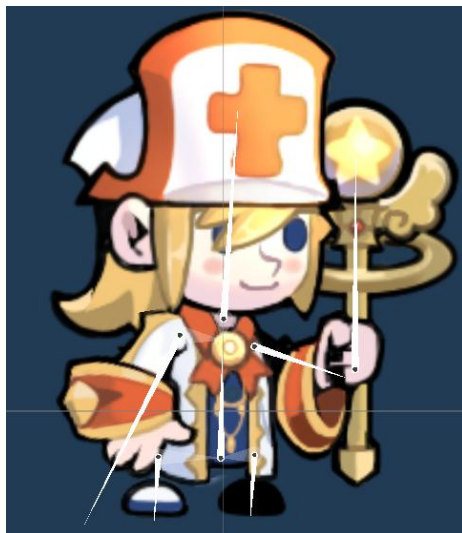


Figure 3: The Healer Wizard.

Monk Wizard

Bio: Focuses on ancient, unhinged magic to perform heroic dark arts to overcome their enemies.

Projectiles: **Cosmicball, Voodoooball**

Playstyle: Strong, mystical spell casting with tide-turning capabilities.

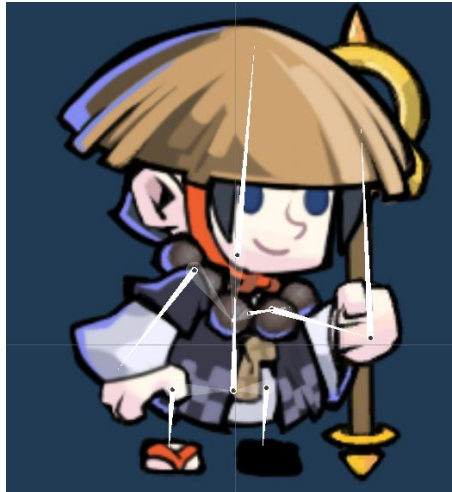


Figure 4: The Monk Wizard.

Enemies (Antagonists)

The forest is defended by three different enemies: **Skeletons**, **Goblins**, and **Golems**, each with three visual variants. The enemies do not move, attack, or perform actions of any sort – they simply serve as destructible targets like many obstacles in the game. However, these are the main target for players to destroy in order to progress.



Figure 5: The Wizard Skeleton, one of three Skeleton variants.

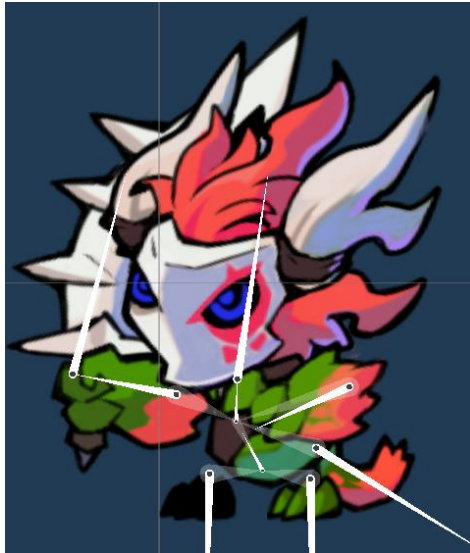


Figure 6: The Berserker Goblin, one of three Goblin variants.



Figure 7: The Ice Golem, one of three Golem variants.

Narrative Flow

The story is narratively delivered subtly through level progression, with each advancing level representing a deeper section of the forest where enemy numbers gradually increase. Upon completion of the final eighth level, players are rewarded with an in-game certificate officially recognizing them as a Guardian of the Forest.

Gameplay Overview

Core Mechanics

Players click and drag their chosen wizard to aim their projectile shot before releasing the mouse button to fire away at the environment. The controls are:

- **Movement & Aiming:** To move the wizard and aim the projectile.
- **Fire:** To fire the projectile.
- **Projectile Power-Up:** To activate the chosen projectile's Power-Up.

Objectives

The one and only objective is to defeat all eight levels of the game. To do so, players must defeat all enemies within the level to advance. To defeat an enemy, players must either hit them directly with their projectile or knock an obstacle onto the enemy's head.

Rules & Systems

Players are limited to three projectiles for the first six levels, and then five projectiles for the final two levels to count for the extra added enemies. Each projectile has its own unique Power-Up that can be activated once per level to aid players in their objective. If players do not defeat all enemies once all their projectiles have been used, they must restart the level.

Projectile	Projectile Behaviour	Power-Up Name	Power-Up Ability
Bloodball	'Bloodies' all objects it touches for its Power-Up combo with average size and speed	Bloodburst	Bursts all bloodied objects at once, sending them flying in random directions
Cosmicball	Super-fast projectile with small size	Cosmic Energy	Enlarges the Cosmicball's size while retaining its super speed
Electricball	Zaps all objects it touches with its big size making up for its slow speed	Zap-Tastic!	Triples the zapping power when electrocuting objects
Fireball	Burns through all objects it touches with high speed	Laser-Beam of Fire	Fires a deadly laser-beam vertically, destroying everything in its path
Poisonball	'Poisons' all objects it touches for its Power-Up combo with average size and speed	Poison Smoke	Destroys all poisoned objects at once

Voodoooball	'Curses' all objects it touches for its Power-Up combo with average size and speed	Way of the Voodoo	Destroys all cursed objects at once with a big explosion and randomly auto-kills a remaining alive enemy
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Progression

Players enter deeper into the forest to continue their mission of clearing all enemies the further they advance for gameplay story progression. Additionally, the final two levels of the game feature a higher enemy count, rewarding players with an additional two projectiles to fire.

Level Design

Level Layout

Each level features a standing platform for the player wizard to stand and move on towards the left-hand side of the screen, while the right-hand side is filled with varied platform designs to hold enemies and obstacles. The environmental structures, enemies, and enemy count vary between levels, while every two levels feature its own unique background and style to represent a different region of the forest.



Figure 8: The level design of Level 4, featuring the player wizard movement platform to the left and the remainder of the level with obstacles and enemies to the right.

Challenges & Puzzles

While the core challenge of the game is physics, the game's level design carefully positions platforms, obstacles, and enemies to encourage strategic projectile choice and aim, as well as smart Power-Up usage. Some projectiles excel in tight spaces, others in destruction, and others in open angles.

Rewards & Progression

The further players advance, the more visual storytelling they discover, as well as an additional two extra projectiles for the final two levels. Upon completion of the game, players are rewarded with a certificate congratulating them on becoming a true Guardian of the Forest.

Art & Visuals

Art Direction

The art direction is a stylized, colourful, cartoon aesthetic which complements the magical and family-friendly theme of the game. This decision allows for approachable visuals that brings the whimsical world to life.

Characters & Environment

Levels feature unique colours and atmospheres tailored to each specific forest region. All three wizards and enemies are from the same asset designer to easily merge them into the world, with unique designs to showcase their personality traits. The environmental background objects and obstacles further enhance the magical forest by seamlessly fitting into the created world with visuals to match.

HUD & UI Elements

The HUD features elements at the top and bottom of the screen to help players with their gameplay. The elements are appropriately sized to ensure gameplay is not affected negatively.

- **Pause/Menu Button (Top-Left):** Pauses the game, displaying the Pause Menu.
- **Level (Top-Right):** Showcases the current level the player is on.
- **Selected Wizard (Bottom-Far Left):** Displays the player's selected wizard.
- **Selected Projectile (Bottom-Middle Left):** Displays the player's selected projectile.
- **Power-Up Activation Status (Bottom-Close Left):** Showcases the projectile's Power-Up activation status. The Power-Up is still available to use if this icon is not greyed out.
- **Remaining Enemies (Bottom-Right):** Shows the remaining enemies alive in the level.



Figure 9: The HUD.

The Main Menu is displayed on a hanging canvas in front of a region of the forest, displaying a variety of elements.

- **Start Button:** For starting a new game.
- **Continue Button:** For continuing from a saved game.
- **Options Button:** For viewing and selecting the wizard and projectile options.
- **Credits Button:** For viewing the credits of the game.
- **Quit Button:** For quitting the game.

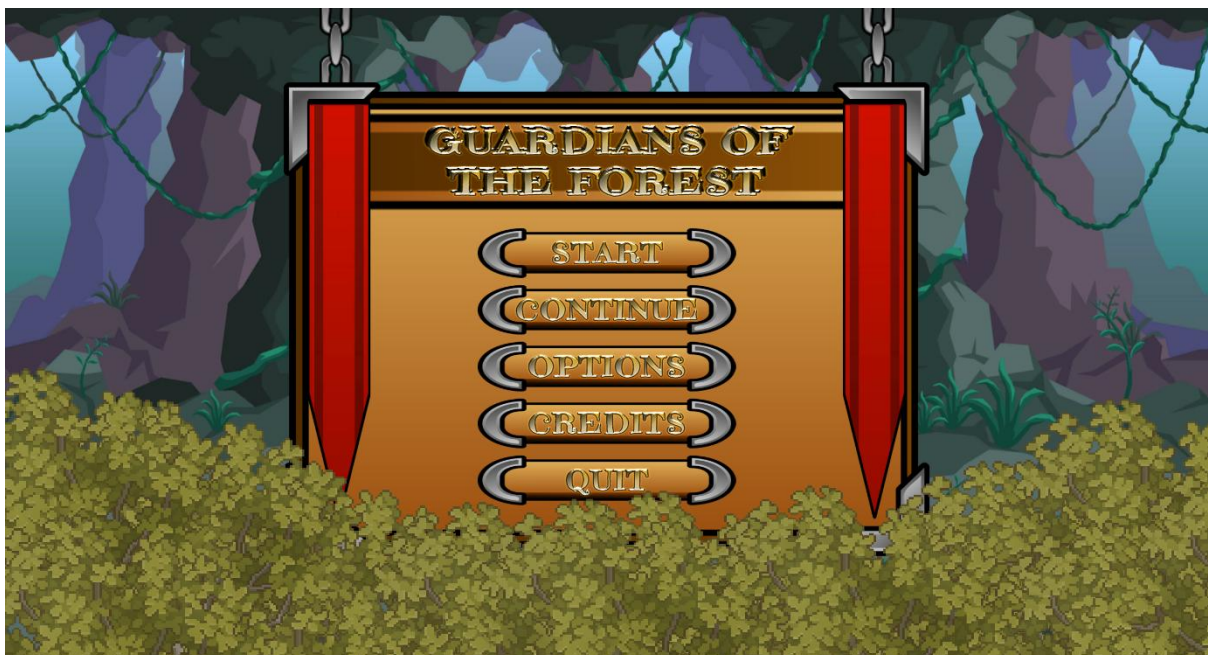


Figure 10: The Main Menu.

The Pause Menu utilizes the same hanging canvas design as the Main Menu, just smaller to partially take up space on the gameplay screen.

- **Resume Button:** Resumes the game.
- **Options Button:** Same function as the Main Menu, minus the ability to select a new wizard and/or projectile.
- **Quit Button:** Quits to the Main Menu.



Figure 11: The Pause Menu.

The Level Failed Menu is similar in design to the Pause Menu, featuring different elements.

- **Continue Button:** Restarts the failed level.

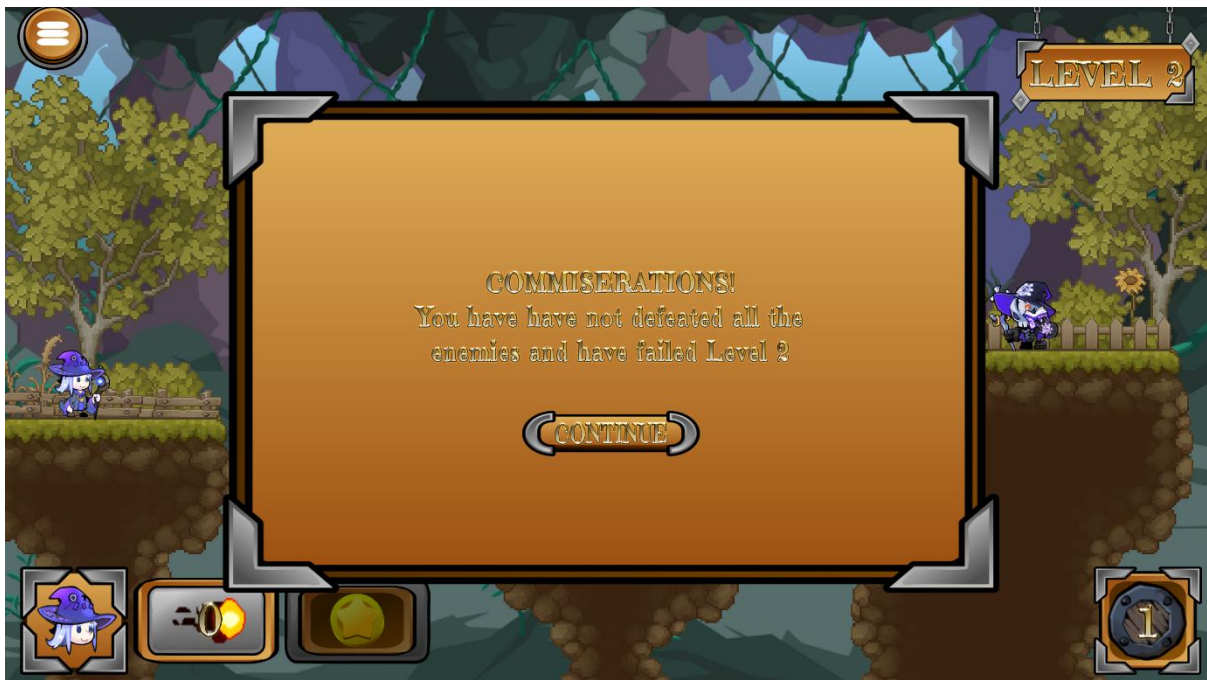


Figure 12: The Level Failed Menu.

The Level Completed Menu is the same as the Level Failed Menu, with its own elements.

- **Continue Button:** Starts the next level.

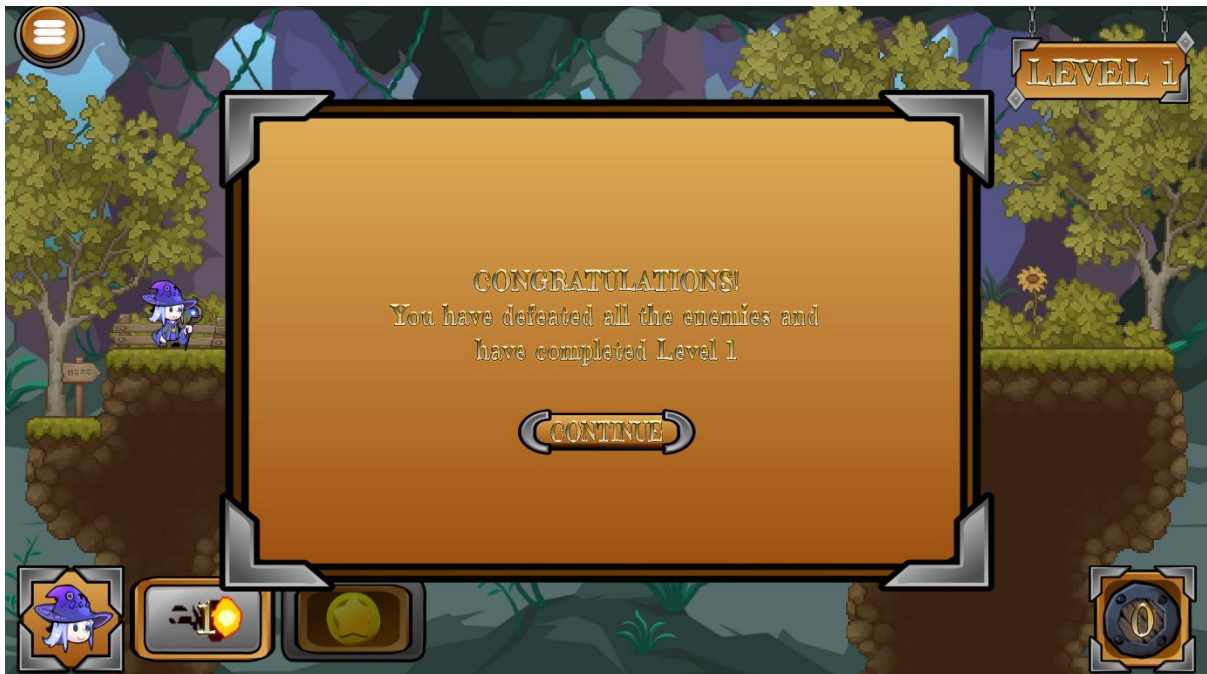


Figure 13: The Level Completed Menu.

The Game Completed Menu also uses the same design as both the Level Failed and Level Completed Menus, containing its own elements.

- **Menu Button:** Returns to the Main Menu.

- **Quit Button:** Quits the game.

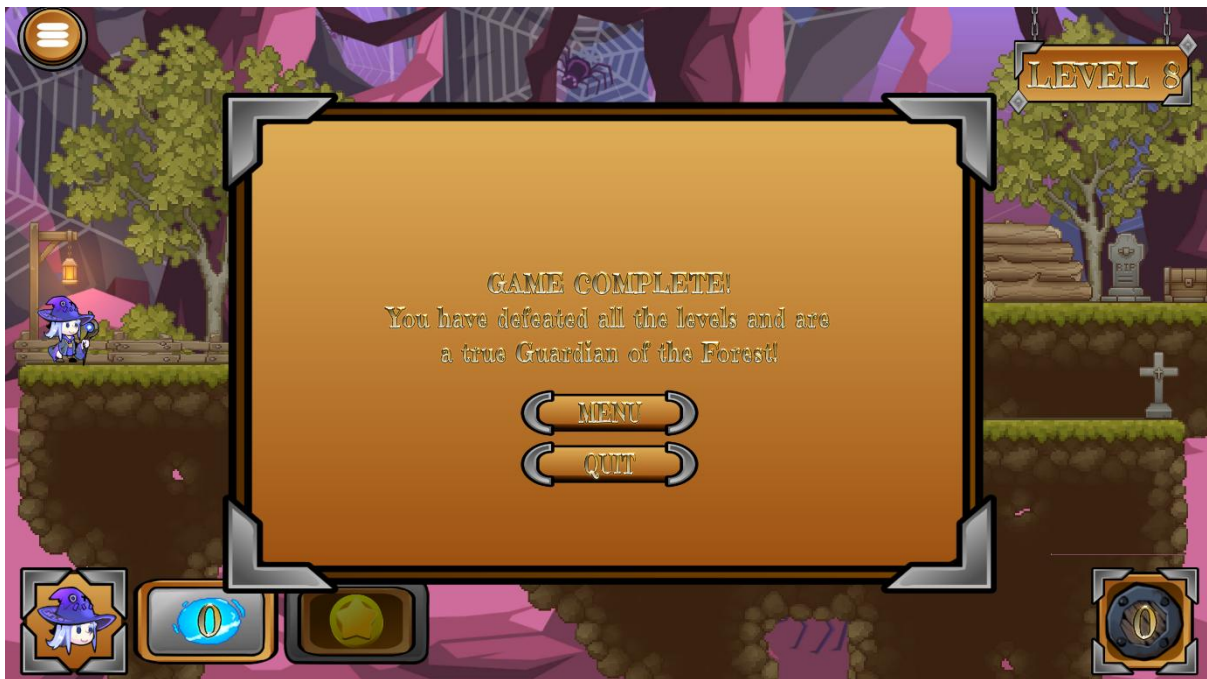


Figure 14: The Game Completed Menu.

The HUD and menus utilize the same one font which was carefully selected to match the theme of the game while also being accessible for all to read with ease.

Animations

Player

Each Guardian Wizard has animations for idling and firing projectiles, each suited to their own personalities to ensure each not only has its own look and gameplay designs, but also its own identity through lifelike movements. Additionally, upon completion of a level, each wizard has its own victory celebration animation.

Projectiles

All six projectiles have their own animations for movement, collision, and Power-Up activation. Each of these animations were carefully selected to bring each projectile to life with their own unique identities, acting as main characters themselves. For example, the Fireball uses a laser-beam of fire animation for its Power-Up, while the Poisonball uses a big poison smoke poof for its Power-Up animation, distinctively separating each projectile.

Some projectile Power-Up animations, such as the Fireball and Voodoo ball, feature multiple visual effects merged into one for a grand design. Although the visuals were sourced from a designer, these specific multi-merge animations were created by myself.

Enemies

All three enemies have their own idling animations shared across their variants. All three enemies share the same death animations due to the functionality being shared and not requiring specific unique animations.

Environment

The environment is full of animated background objects and obstacles, such as fragile wooden planks, boxes, chests, campfires, and more. These help bring the world and the game to life with a 'real feel', further helping to enhance the magical effect of the game.

UI

All menus use a handcrafted animation which sees the hanging canvas drop down from the top of the screen on chains to add more magic and personality to the overall feel of the game.

Audio

Music

Guardians of the Forest is an upbeat, casual, fantasy themed game, and therefore uses catchy soundtracks for each individual level and menu. Each level progressively gets harder as the player nears the more dangerous regions of the forest, which is not only shown by the scenery and enemy count, but also through the soundtracks which progressively get more intense.

The menus ensure players feel comfortable and relaxed with charming fantasy-themed music appropriate for menu navigation.

Sound Effects

There are a handful of sound effects featured in *Guardians of the Forest* to further help bring the magical world to life. The wizards all share the same firing sound when a projectile is fired, and each enemy type uses its own idling sound to audibly show they are alive – until defeated, where a distinct death sound and poof sound are played one after another.

All six projectiles have their own unique sounds for movement, collision, and Power-Up activation to provide important audio cues for players. All UI elements utilize appropriate button click sounds for when buttons are clicked.

These sound effects help glue the game together with a polished design to ensure players feel as immersed as possible within the experience.

Controls & User Interface

Controls

Movement & Aiming (Left-Mouse Click)

Allows the player to move their character and aim the projectile.

Fire (Left-Mouse Unclick)

Allows the player to fire their projectile.

Projectile Power-Up (Enter)

Allows the player to activate their projectile's Power-Up.

A key rebinding system is not implemented due to the casual, easy-to-play nature of the game and controls.

UI Flow

The HUD can feature a handful of elements at any given time, but they are all strategically sized and placed to avoid interruption with gameplay. To ensure consistency across all HUD, menus, and other UI elements, the same 'hanging canvas' design is used. All menus support full mouse navigation for ease-of-use.

Technical Specifications

Engine & Tools

Guardians of the Forest was developed with Unity and C# for core systems. Visual Studio 2022 (coding) and Audacity (audio) was used for other purposes.

Platform Requirements

Minimum Requirements:

- **OS:** Windows 10 64-bit
- **Processor:** Intel Core i5-4th Gen or equivalent
- **Memory:** 4 GB RAM
- **Graphics:** NVIDIA GeForce GTX 750 Ti / AMD Radeon R7 260X
- **VRAM:** 1 GB
- **Storage:** 200 MB available space
- **DirectX:** Version 11

Recommended Requirements:

- **OS:** Windows 10/11 64-bit
- **Processor:** Intel Core i7-9th Gen or equivalent
- **Memory:** 8 GB RAM
- **Graphics:** NVIDIA GeForce GTX 1060 / AMD Radeon RX 580
- **VRAM:** 3 GB
- **Storage:** 500 MB available space
- **DirectX:** Version 12

Performance Targets

- **Frame Rate:** 60 FPS locked
- **Resolution:** 1920x1080, adapts to player's display resolution
- **Loading Times:** ~1-2 seconds on SSD; ~5-10 seconds on HDD

Development & Testing Hardware

Primary Development Machine

- **OS:** Windows 10 64-bit
- **Processor:** Intel Core i7-9750H
- **Memory:** 16 GB RAM
- **Graphics:** NVIDIA GeForce RTX 2080 Super Max-Q
- **VRAM:** 8 GB
- **DirectX:** Version 12

Quality Assurance Testing

The game was tested on the primary development machine only due to the purpose of the game being to create a small scope starter project for my learning. I, amongst a few close others, helped test the functionality, performance, and overall fun-factor to ensure the best experience was created.

Postmortem

What Went Well

- Many skills from learning the Unity game engine to programming practices not only with the C# language but also with general practices that can be used across various other languages were successfully learned throughout the project's creation, which was one of the goals of the project.
- The game works as intended with smooth, enjoyable gameplay and a unique, fantasy theme bringing the entire project together.
- If the project was to be revisited for additional work, a solid foundation has been created to allow this.

What Could Have Been Better

- The project began with minimal pre-production planning, with only the idea of the game being established without any proper plan to implement the project. Much of the project was made reactively rather than intentionally, which didn't affect the production much due to the game's small scope, but could have affected it more if the game was larger.
- The game was tested only on the primary development machine and by a small group of people, meaning the breadth of user testing was limited which therefore limited the amount of feedback received.
- The use of version control with Git was at times minimal and inconsistent due to the project being a first solo development, where knowledge of version control was limited. Taking the time to learn more about this pre-production would have resulted in better version control management – which is crucial to a game's development cycle.
- Some features took a while to implement due to lack of experience. Better scheduling would have helped keep development momentum while managing expectations.

Key Lessons Learned

- The importance of the pre-production phase of development is not to be underestimated. Creating documentation, prototypes, technical plans, and learning necessary systems beforehand dramatically improves efficiency, clarity, and reduces redesign work later.
- Building small and functional systems then iterating on them proved far more effective than building large interconnected systems all at once.
- Tracking workload and setting realistic goals keeps the project's scope small and achievable.

- Completing a game solo – from programming to audio manipulation to asset acquisition – provided hands-on experience across the entire development pipeline.

Future Plans

Guardians of the Forest is a standalone experience with no plans for any downloadable content or future sequels. However, the game could easily be expanded upon due to the solid framework created, where a variety of new levels, wizards, projectiles, and enemies could be implemented if any thoughts of a return were to occur.

References & Inspirations

Games

The main inspiration for *Guardians of the Forest* is, of course, the *Angry Birds* franchise. The game uses the same mechanics and gameplay themes as *Angry Birds*, just with its own twist on setting, story, characters, and some minor mechanical changes.

Other

Fantasy and magic themes are heavily present throughout the game. These ideas were chosen as they are widely regarded as family-friendly and help bring imagination to life in fun and creative ways with minimal restrictions.